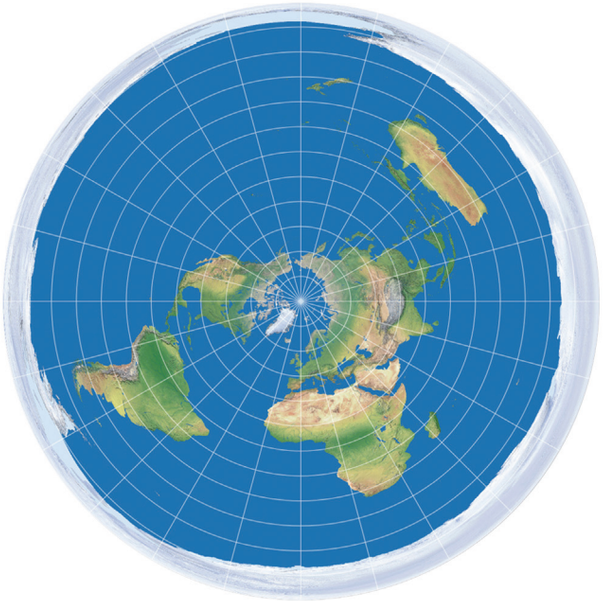
Polar Zenithal Equi-distant Projection



Properties of Polar Zenithal Equi distant Projection

- The pole is a point forming the centre of the projection and the parallels are concentric circles.
- The meridians are straight lines radiating from pole having correct angular distance between them.
- The meridians intersect the parallels at right angles.
- The spacing between the parallels represent true distances, therefore the scale along the meridians is correct.
- The scale along the parallels increases away from the centre of the projection.
- It is used for preparing maps of polar areas for general purposes.

Example 10.4

Construct a Polar zenithal equi-distant projection for the whole globe with a reduced earth radius of 3cm and the latitudinal and longitudinal interval 30°.

Steps of construction:

- Draw a circle with radius equal to 3cm representing a globe with O as centre.
- Mark the angle of 30° and label it as AOB.
- With radius equal to AB as centre draw a circle with centre N.
- Let CD and AB be the polar and equatorial diameters respectively which intersect each other at right angles at O, the centre of the circle.
- The number of intervals will be 90/30 = 3.
- Draw 3 concentric circles with N as centre. Mark the meridians radiating from the centre N.
- Using protractor, draw other radii at 30° interval to represent other meridians.
- Complete the projection as shown in Figure.

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